

MARCO CHEN

+1 (548) 577-5684 | z253chen@uwaterloo.ca | [Github](#) | [LinkedIn](#) | [Website](#)

EDUCATION

University of Waterloo

Bachelor of Computer Science, Co-op

Sep. 2023 – Aug. 2028

CGPA: 3.7/4.0

- **Coursework:** OOP, DSA, Operating Systems, Compilers, Graph Theory, Probability, Statistics, Numerical Methods

EXPERIENCE

Team Lead, Machine Learning Engineer

WAT.ai | Python, PyTorch, NumPy, CUDA

Jun. 2026 – Current

Waterloo, ON, Canada

- Lead a 13-student research team developing a world model for multi-agent soccer dynamics to forecast latent states and simulate imagined trajectory rollouts for decision-making and planning.
- Architected a State Space Model (SSM) in PyTorch/CUDA to compress high-dimensional positional tracks into compact latent representations, capturing spatio-temporal player interactions.
- Designed an end-to-end data ingestion pipeline to extract per-object coordinate data from raw broadcast video.

Software Engineer Intern

Geotab | C#, TypeScript, PostgreSQL, Google BigQuery, React, .NET

May 2026 – Aug. 2026

Kitchener, ON, Canada

- Optimized harsh driving event recommendation load latency by parallelizing API calls and refactoring slow database queries, reduced vehicle address reverse geocoding latency by 90% and total request latency by 75%.
- Implemented a data persistence layer and an offline-first synchronization module with a local message queue for a telematics mobile app, caching user activities and auto-syncing payloads upon network reconnection, enabling full offline access and cutting redundant API calls by 40%.
- Built an end-to-end error observability dashboard, efficiently processing 23M+ rows and routing 565K+ errors.

Software Engineer Intern

Geotab | C#, TypeScript, PostgreSQL, xUnit, React, .NET

Sep. 2025 – Dec. 2025

Kitchener, ON, Canada

- Optimized and refactored service initialization and synchronization logic, reducing JSON-RPC API calls by 46%.
- Led the automation of the Transport Canada ELD test procedures, resulting in a 98% reduction in execution time.
- Investigated and debugged backward compatibility bugs, improving consistency across different OS platforms.

Software Engineer Intern

Octopodi Technologies | Rust, TypeScript, SQLite, React, Next.js, Tauri

Jan. 2025 – Apr. 2025

Waterloo, ON, Canada

- Engineered a fully local, cross-platform plagiarism detection tool using Tauri, Rust, and Next.js, executing AST-based pattern matching and tokenization to analyze codebases with zero cloud dependencies.
- Implemented an asynchronous Tauri IPC bridge and SQLite persistence layer to cache indexed code footprints.

PROJECTS

LLM From Scratch | Python, PyTorch, WandB, Multiprocessing, einops

- Built an optimized BPE tokenizer with a max-heap and multiprocessing, reducing merge complexity to $O(n \log m)$.
- Implemented RMSNorm, SwiGLU FFN, RoPE, and scaled dot-product Multi-Head Attention from scratch, trained a 22M Transformer to 2.20 val perplexity and 0.78 val loss on TinyStories.
- Developed a training pipeline using BF16 mixed-precision, zero-copy strided batching, custom AdamW optimizer, cosine annealing, and gradient clipping, sustaining 413k tokens/sec throughput verified on an H100 GPU.

Mini-vLLM | Python, PyTorch, MLX, PagedAttention

- Designed a minimal implementation of vLLM, optimizing inference performance for Apple Silicon chips using MLX.
- Implemented PagedAttention, Continuous Batching, Chunked-Prefill, and Prefix Caching, achieving 5.4x higher decode throughput than single-stream decode on a base M4 chip.

TECHNICAL SKILLS

Languages: Python, C++, Go, TypeScript, C#, SQL

Frameworks: React, Next.js, FastAPI, Flask, Django, chi

AI/ML: PyTorch, vLLM, SGLang, HuggingFace, LangGraph, NumPy, TensorFlow, Keras, MLX, OpenCV, WandB

Tools & Platforms: Git, Linux, Unix, Bash, CUDA, Docker, Kubernetes, Prometheus, Grafana, GCP, AWS, PostgreSQL, Redis, Kafka